

TABLEAU FOR BEGINNERS

Module 9 :- Advanced Tableau Features

Action Filters • LOD Expressions • Drill Down Reports

Topics in This Module

Topic 1	Action Filters — Make charts talk to each other with one click
Topic 2	LOD Expressions — Calculate at any level you choose, not just what is on screen
Topic 3	Drill Down Reports — Let users explore data from summary to detail interactively

Dataset: Online Store Customer Orders • 20 Rows • Step-by-Step Every Feature

Our Dataset — Online Store Customer Orders

This dataset has 20 orders from an online store. Each row is one order placed by a customer. We have 8 customers across 4 regions, 3 product categories, and 6 months of data. This structure is perfect for Action Filters, LOD Expressions, and Drill Down Reports.

Full Dataset — Customer Orders

Order_I D	Customer	Regio n	City	Category	Product	Order_Da te	Sales (₹)	Profit (₹)	Qty
ORD001	Rahul Sharma	North	Delhi	Electronics	Laptop	05-Jan-2024	55,000	12,000	1
ORD002	Priya Patel	West	Mumbai	Clothing	T-Shirt	08-Jan-2024	1,200	400	3
ORD003	Amit Verma	South	Chennai	Books	Data Science Book	12-Jan-2024	800	200	2
ORD004	Rahul Sharma	North	Delhi	Electronics	Mobile Phone	15-Jan-2024	18,000	4,200	1
ORD005	Sneha Singh	East	Kolkata	Clothing	Jeans	20-Jan-2024	2,400	700	2
ORD006	Priya Patel	West	Mumbai	Electronics	Tablet	25-Jan-2024	24,000	5,500	1
ORD007	Kiran Kumar	South	Bangalore	Books	Python Book	28-Jan-2024	600	150	1
ORD008	Amit Verma	South	Chennai	Clothing	Jacket	02-Feb-2024	3,500	900	1
ORD009	Rahul Sharma	North	Delhi	Books	Excel Book	05-Feb-2024	500	120	1
ORD010	Deepa Nair	West	Pune	Electronics	Headphones	10-Feb-2024	4,200	950	2
ORD011	Sneha Singh	East	Kolkata	Electronics	Mobile Phone	14-Feb-2024	17,500	4,000	1
ORD012	Vikram Mehta	North	Delhi	Clothing	Formal Shirt	18-Feb-2024	1,800	500	2
ORD013	Kiran Kumar	South	Bangalore	Electronics	Tablet	22-Feb-2024	22,000	5,000	1
ORD014	Deepa Nair	West	Pune	Clothing	Kurta	01-Mar-2024	1,500	420	3
ORD015	Priya Patel	West	Mumbai	Books	ML Book	05-Mar-2024	950	240	1
ORD016	Vikram Mehta	North	Delhi	Electronics	Laptop	10-Mar-2024	52,000	11,500	1

Order_ID	Customer	Region	City	Category	Product	Order_Date	Sales (₹)	Profit (₹)	Qty
ORD017	Amit Verma	South	Chennai	Electronics	Headphones	15-Mar-2024	3,800	850	1
ORD018	Rahul Sharma	North	Delhi	Clothing	Jacket	20-Mar-2024	3,200	880	1
ORD019	Sneha Singh	East	Kolkata	Books	Statistics Book	25-Mar-2024	700	180	2
ORD020	Kiran Kumar	South	Bangalore	Clothing	T-Shirt	30-Mar-2024	1,100	320	2

Key Points About This Dataset

8 Customers: Rahul Sharma, Priya Patel, Amit Verma, Sneha Singh, Kiran Kumar, Deepa Nair, Vikram Mehta.

4 Regions: North (Delhi), West (Mumbai, Pune), South (Chennai, Bangalore), East (Kolkata).

3 Categories: Electronics (high value), Clothing (medium), Books (low value).

Notice: Rahul Sharma placed 4 orders. Priya placed 3 orders. Others placed 2-3 each.

This multiple-orders-per-customer structure is exactly what makes LOD expressions powerful.

Topic 1 — Action Filters



Action Filters

Click on one chart and another chart automatically updates — charts talk to each other

1.1 What is an Action Filter? — Very Simple Explanation

An Action Filter is a connection between two charts on a dashboard. When you click on something in Chart A — a bar, a dot, a row — Chart B automatically filters to show only the related data. It is like clicking a link on a webpage that takes you to a new page with more detail.

Real-life example: *You have a dashboard with two charts: a bar chart showing sales by Region (top), and a table showing all orders (bottom). Without Action Filters, clicking the North bar does nothing to the table. With an Action Filter, clicking 'North' on the bar chart makes the table automatically show only North region orders. You do not need to add a separate filter — the click does it.*

Why Action Filters Are So Powerful

They turn a static dashboard into an interactive exploration tool. Users can investigate data at their own pace — click a region, see its orders. Click a customer, see all their purchases. Click a category, see the trend for just that category. All without any dropdown filters cluttering the screen.

1.2 The 3 Types of Actions

Action Type	What It Does	Simple Example
Filter Action	Clicking something on one chart FILTERS another chart to show only matching data	Click 'North' bar → order table shows only North orders
Highlight Action	Clicking something HIGHLIGHTS matching marks across all charts — does not hide non-matching data, just dims it	Click 'Electronics' → all Electronics marks glow, others fade
URL Action	Clicking something OPENS a website or URL in a browser	Click a customer name → opens their profile page on your website

1.3 Step-by-Step — Create a Filter Action

We will build a two-chart dashboard: a bar chart of Sales by Region (master chart) and a table of all orders (detail chart). Clicking a region bar will filter the order table.

Part A — Build the Two Charts First

1	Open Tableau and connect to the Customer Orders dataset
2	On Sheet 1 — build the Master Chart (Bar Chart): Drag 'Region' to Rows shelf Drag 'Sales (₹)' to Columns shelf Sort descending. Rename sheet: 'Sales by Region'
3	On Sheet 2 — build the Detail Chart (Order Table): Drag 'Order_ID' to Rows shelf Drag 'Customer' to Rows shelf (next to Order_ID) Drag 'Product' to Rows shelf Drag 'Sales (₹)' to Rows shelf Rename sheet: 'Order Details'
4	Create a new Dashboard: click the '+' icon at the bottom → New Dashboard
5	From the left panel, drag 'Sales by Region' to the dashboard canvas — it fills the top half
6	Drag 'Order Details' to the bottom half of the dashboard
7	You now have both charts on the same dashboard screen

Part B — Add the Filter Action

8	With the dashboard open, go to the top menu: Dashboard → Actions
9	A dialog box opens showing all existing actions (probably empty). Click 'Add Action' → Filter
10	The 'Add Filter Action' dialog opens. Fill it in: → Name: type 'Region Click Filter' (any name you like) → Source Sheets: check 'Sales by Region' (this is the chart user will click on) → Run Action On: select 'Select' (action runs on click) → Target Sheets: check 'Order Details' (this chart will be filtered)
11	Under 'Clearing the Selection', choose 'Show All Values' (This means: when user clicks blank area to deselect, show all orders again)
12	Click OK → click OK again to close the Actions dialog
13	TEST IT: Click the 'North' bar on the bar chart The Order Details table below should instantly show only ORD001, ORD004, ORD009, ORD012, ORD016, ORD018 (all North/Delhi orders)
14	Click the 'West' bar → table shows Priya Patel and Deepa Nair's orders only

15

Click on blank canvas (not on any bar) → all orders come back

1.4 Step-by-Step — Create a Highlight Action

A Highlight Action does NOT hide other data — it just makes matching marks brighter and dims everything else. Good when you want to show context but still see the full picture.

1	On your dashboard, go to Dashboard → Actions → Add Action → Highlight
2	Fill in: → Name: 'Category Highlight' → Source Sheets: 'Sales by Region' → Run Action On: 'Hover' (highlight happens when mouse hovers, not click) → Target Sheets: 'Order Details' → Target Highlighting: Selected Fields → check 'Category'
3	Click OK
4	TEST IT: Hover your mouse slowly over the 'North' bar In the Order Details table, the rows with Electronics orders will glow — others fade Move mouse away — everything returns to normal brightness

1.5 Filter vs Highlight — When to Use Which

Filter Action	Highlight Action
HIDES non-matching data	DIMS non-matching data — still visible
Run on: Click (select)	Run on: Hover is most natural
Best for: Master-detail navigation	Best for: Cross-highlighting patterns
User drills INTO detail	User sees context without losing overview
Click region → only that region's orders	Hover region → matching orders glow

1.6 Step-by-Step — Create a URL Action

A URL Action opens a website when a user clicks on data. Useful for linking to product pages, customer profiles, or external reports.

1	On your dashboard, go to Dashboard → Actions → Add Action → Go to URL
2	Fill in: → Name: 'Google Customer Name' → Source Sheets: 'Order Details' → Run Action On: Menu (a right-click menu appears)
3	In the URL box, type: <a href="https://www.google.com/search?q=<Customer>">https://www.google.com/search?q=<Customer> The <Customer> part is a field reference — Tableau replaces it with the actual customer name when clicked
4	Click OK
5	TEST IT: Right-click any row in the Order Details table A menu appears with 'Google Customer Name' Click it — a browser window opens and searches for that customer's name on Google

Topic 2 — Level of Detail (LOD) Expressions



LOD Expressions

Calculate at any level you want — not just what is currently shown on your chart

2.1 The Problem LOD Solves — A Very Simple Story

Before learning LOD, you need to understand the problem it solves. Here is a common situation that confuses beginners:

The Problem

You want to show: each order's sales AND the customer's TOTAL sales across all orders — side by side.

For example, Rahul Sharma placed 4 orders (ORD001, ORD004, ORD009, ORD018).
His total sales across all 4 orders = ₹55,000 + ₹18,000 + ₹500 + ₹3,200 = ₹76,700.

If you just drag Customer and Sales to the view, Tableau shows ₹76,700 as one row.
But if you add Order_ID to break it into individual orders, Tableau shows 4 rows of ₹55,000, ₹18,000, ₹500, ₹3,200.

Now you LOST the total ₹76,700 — it disappeared because Tableau calculated at Order level.

LOD fixes this: it lets you calculate the TOTAL at Customer level even when the view is at Order level.

Simple analogy: *You are looking at individual test scores for each student in each subject. You also want to show each student's TOTAL marks across all subjects next to every row. Normal aggregation gives you one or the other. LOD gives you both at the same time.*

2.2 What Does LOD Stand For?

LOD = Level of Detail. It means: at what level should Tableau do the calculation?

LOD Type	Keyword	Simple Meaning	When to Use
FIXED LOD	FIXED	Calculate at the level I specify — ignore everything else on the view	Total sales per customer, regardless of what filters or dimensions are on the chart
INCLUDE LOD	INCLUDE	Calculate at a MORE detailed level than what is on the view	Average sales per order per customer — more detail than just per customer
EXCLUDE LOD	EXCLUDE	Calculate at a LESS detailed level than what is on the view	Region-level average — even if City is shown on the chart

2.3 FIXED LOD — The Most Common and Most Useful

FIXED LOD calculates at a specific level that YOU define — and it stays fixed even when the view changes or when filters are applied.

Simple Definition of FIXED

FIXED says: 'I want to calculate this measure grouped by THIS field — no matter what else is on the chart.' It locks the calculation to a specific level.

Syntax:

```
{ FIXED [dimension] : aggregation([measure]) }
```

Example:

```
{ FIXED [Customer] : SUM([Sales (₹)]) }
```

This means: for EACH unique Customer, sum up all their Sales.

This gives one total per customer — regardless of what other dimensions are on the chart.

Practical Example — Customer Total Sales alongside each Order:

Order_ID	Customer	Order Sales (₹)	Customer Total Sales (LOD)	% of Customer Total
ORD001	Rahul Sharma	55,000	76,700	71.7%
ORD004	Rahul Sharma	18,000	76,700	23.5%
ORD009	Rahul Sharma	500	76,700	0.7%
ORD018	Rahul Sharma	3,200	76,700	4.2%
ORD002	Priya Patel	1,200	26,150	4.6%
ORD006	Priya Patel	24,000	26,150	91.8%
ORD015	Priya Patel	950	26,150	3.6%

The 'Customer Total Sales' column shows the same value (₹76,700) for all Rahul's rows — that is FIXED LOD working. It does not change per row.

Step-by-Step — Create FIXED LOD

1

Go to Analysis → Create Calculated Field

2	Name it: 'Customer Total Sales (LOD)'
3	Type the formula exactly: ,
4	The curly braces { } are required — they tell Tableau this is an LOD expression
5	Check the bottom: green tick should say 'The calculation is valid'
6	Click OK — the new field appears in the Data panel with a small '=' icon
7	Build a view: drag Order_ID to Rows, Customer to Rows, Sales to Rows
8	Drag 'Customer Total Sales (LOD)' also to Rows
9	You now see: each order's individual sales AND the customer's total alongside — all in one table
10	Try adding a filter — filter to show only Electronics orders Notice: the Customer Total Sales (LOD) still shows ₹76,700 for Rahul This is because FIXED ignores filters (this is both its power and something to be careful about)

2.4 INCLUDE LOD — Add More Detail Than the View

INCLUDE adds a dimension to the calculation that is NOT on the current view. It calculates at a MORE granular level, then aggregates up for the chart.

Simple Definition of INCLUDE

INCLUDE says: 'Add this extra dimension to the calculation — even if it is not visible on the chart.' The result is then averaged or summed up to match the view level.

Example — Average Sales per Order per Customer:

If you just do AVG(Sales) at the Customer level, you get the average across all orders. With INCLUDE, you can first calculate the total per order, then average those order totals per customer.

```
{ INCLUDE [Order_ID] : SUM([Sales (₹)]) }
```

This tells Tableau:

Step 1 → Include Order_ID in the calculation: sum sales per Order_ID

Step 2 → The view then aggregates these order-level totals (e.g. AVG them per Customer)

```
Use it as: AVG( { INCLUDE [Order_ID] : SUM([Sales (₹)]) } )  
Result: average order value per customer
```

1	Go to Analysis → Create Calculated Field
2	Name it: 'Avg Order Value per Customer (LOD)'
3	Type: AVG({ INCLUDE [Order_ID] : SUM([Sales (₹)]) })
4	Click OK
5	Drag 'Customer' to Rows
6	Drag your new LOD field to Columns — you see average order value per customer
7	Compare with just AVG([Sales]) — the results differ because INCLUDE goes deeper before averaging

2.5 EXCLUDE LOD — Remove Detail From the Calculation

EXCLUDE removes a dimension from the calculation — even if that dimension IS on the current view. It calculates at a higher (less detailed) level.

Simple Definition of EXCLUDE

EXCLUDE says: 'Ignore this dimension in the calculation — even if it is currently showing on the chart.' Useful when you want to show a regional average on a city-level chart.

Example — Show Region-level average sales even when City is on the chart:

```
{ EXCLUDE [City] : AVG([Sales (₹)]) }
```

Scenario: Your chart has Region AND City on the Rows shelf.

Normally, average sales would calculate per City.

With EXCLUDE [City], it calculates average sales per Region only — ignoring City even though City is visible on the chart.

Result: Every city within North shows the same value — the North Region average (₹25,125)

This lets you compare each city against its Region average in the same view.

1	Build a view: Region and City on Rows, AVG(Sales) on Columns
2	You see average sales per City: Delhi ₹21,950, Mumbai ₹8,717, etc.
3	Create Calculated Field: 'Region Avg Sales (LOD)' Formula: { EXCLUDE [City] : AVG([Sales (₹)]) }
4	Drag this new field also to the Columns shelf (dual axis or second bar)
5	Now each row shows: the City's own average AND the Region average side by side
6	This immediately tells you: is this city above or below its region's average?

2.6 FIXED vs INCLUDE vs EXCLUDE — One-Line Summary

LOD Type	One-Line Rule	Quick Memory Trick
FIXED	Lock to a specific level — ignore the view	I DECIDE the level, not Tableau
INCLUDE	Go deeper than the view — add more detail	Include MORE detail in the calc
EXCLUDE	Go higher than the view — remove some detail	Exclude a dimension, zoom OUT

Important — FIXED and Filters

By default, FIXED LOD ignores dimension filters on the view. This means if you filter to show only Electronics, the FIXED customer total still includes ALL orders including Clothing and Books.

To make FIXED respect a filter: drag that filter field to the Context Filter (right-click the filter pill → Add to Context). Context Filters run before LOD calculations.

Topic 3 — Drill Down Reports



Drill Down Reports

Start with a summary view — click to go deeper — click to go back up

3.1 What is a Drill Down Report? — Very Simple Explanation

A Drill Down Report starts at a high level (big picture) and lets the user click to go deeper into the details — level by level. The user controls how deep they want to go. When they have seen enough detail, they click to go back up.

Real-life example: *On Amazon, you start at the homepage (all products). You click 'Electronics' — you drill down one level. You click 'Laptops' — another level deeper. You click a specific laptop model — the deepest level showing full product details. At each step you clicked to go deeper. That is drill down.*

Our Drill Down Path for This Module

Level 1 → Region: North, South, West, East (the big picture — 4 bars)

Level 2 → City: Delhi, Mumbai, Chennai, Bangalore, Kolkata, Pune (drill into a region)

Level 3 → Customer: Rahul Sharma, Priya Patel, etc. (drill into a city)

Level 4 → Product: Laptop, Mobile Phone, etc. (the finest detail level)

3.2 Method 1 — Hierarchy Drill Down (Using + and – Buttons)

This is the simplest method — build a hierarchy (like we did in Module 5) and Tableau automatically gives you + and – buttons on the chart to expand and collapse levels.

Step-by-Step — Build the Hierarchy

1	In the Data panel, find the 'Product' field
2	Drag 'Product' and drop it directly ON TOP of 'Category' in the Data panel A dialog asks: 'Create Hierarchy?' → Click OK Name it: 'Product Hierarchy'
3	Now drag 'Customer' and drop it ON TOP of 'Category' inside the hierarchy (Customer becomes a level between Category and Product)
4	Drag 'City' and drop it ON TOP of 'Customer' inside the hierarchy
5	Drag 'Region' and drop it ON TOP of 'City' inside the hierarchy

6	Your hierarchy now shows (top to bottom): Region → City → Customer → Category → Product
7	If the order is wrong, drag levels up or down inside the hierarchy to fix the order

Step-by-Step — Use the Hierarchy in a Chart

8	Open a new blank worksheet
9	Drag 'Region' (from inside the Product Hierarchy) to the Rows shelf You see 4 rows: East, North, South, West
10	Drag 'SUM(Sales)' to the Columns shelf — 4 horizontal bars appear
11	Look at the chart: next to each Region name you see a small '+' icon
12	Click the '+' next to 'North' North expands to show its cities: Delhi
13	Click '+' next to 'Delhi' Delhi expands to show its customers: Rahul Sharma, Vikram Mehta
14	Click '+' next to 'Rahul Sharma' Expands to show Rahul's categories: Books, Clothing, Electronics
15	Click '+' next to 'Electronics' to see individual products: Laptop, Mobile Phone
16	Click '-' on any level to COLLAPSE it back up to the parent
17	Right-click any '+' → 'Expand All' to expand all members of that level at once

3.3 Method 2 — Action-Based Drill Down (Click Chart to Navigate)

This is a more advanced drill down. Instead of + and – buttons, the user clicks on a bar and it completely replaces the chart with the next level. More like a real application — clean and professional.

What we will build:

- Chart 1 — Region Summary: 4 bars, one per region (the starting view)
- Chart 2 — City Detail: shows cities filtered to the region the user clicked
- Chart 3 — Customer Detail: shows customers filtered to the city the user clicked
- A 'Back' button on each chart to return to the previous level

Part A — Build the 3 Level Charts

1	Sheet 1 — 'Region Level': Region → Rows, SUM(Sales) → Columns, sorted descending Rename sheet: 'L1 Region'
2	Sheet 2 — 'City Level': City → Rows, SUM(Sales) → Columns, sorted descending Rename sheet: 'L2 City'
3	Sheet 3 — 'Customer Level': Customer → Rows, SUM(Sales) → Columns, Category → Color Rename sheet: 'L3 Customer'

Part B — Build the Dashboard with Hidden Sheets

4	Create a new Dashboard
5	Drag 'L1 Region' chart to fill the whole dashboard canvas (This is the default starting view)
6	Drag 'L2 City' to the dashboard as well — but drag it to the same area so it overlaps L1 A layout container will be created — both sheets sit in the same space
7	Drag 'L3 Customer' also to the same container
8	Click on the L2 City sheet in the dashboard → look for the small grey arrow at the top right Click it → select 'Hide' L2 City is now hidden — invisible on the dashboard
9	Do the same for L3 Customer — hide it
10	Now only L1 Region is visible — the starting state of your drill down report

Part C — Add Filter Actions to Connect the Levels

11	Go to Dashboard → Actions → Add Action → Filter
12	Action 1 — Region to City: → Name: 'Drill to City' → Source: 'L1 Region' Run On: Select → Target: 'L2 City' Click OK
13	Action 2 — City to Customer: → Name: 'Drill to Customer' → Source: 'L2 City' Run On: Select → Target: 'L3 Customer'

	Click OK
14	Now add Show/Hide Button Actions: Dashboard menu → Objects → drag a 'Show/Hide Button' object onto the dashboard
15	Configure Show/Hide Button: → Click the button → Item to Show/Hide: L2 City → Show Button Text: 'View by City' → This button will show the City chart and hide the Region chart when clicked
16	Add another Show/Hide button for L3 Customer
17	Add a 'Back' navigation button: Dashboard → Actions → Add Action → Filter → Source: L2 City → Clearing Selection: Show all values → this acts as going back

3.4 Simple Alternative — Use a Parameter for Drill Down

If the Show/Hide method feels complex, here is a simpler approach using a Parameter to control which level is shown.

1	Create a Parameter: Name = 'Drill Level', Type = Integer, Values = 1, 2, 3 (1 = Region, 2 = City, 3 = Customer)
2	Create a Calculated Field: 'Drill Dimension' IF [Drill Level] = 1 THEN [Region] ELSEIF [Drill Level] = 2 THEN [City] ELSE [Customer] END
3	Build one chart: drag 'Drill Dimension' to Rows, Sales to Columns
4	Show the Parameter Control: right-click parameter → Show Parameter Control
5	A selector appears on the dashboard: 1, 2, or 3
6	When user selects 1 → chart shows Regions When user selects 2 → chart shows Cities When user selects 3 → chart shows Customers
7	Combine with an Action Filter from a Region chart to auto-set the parameter level — full drill down in one chart

3.5 Drill Down vs Filter — What is the Difference?

Regular Filter	Drill Down
User manually selects from a dropdown	User clicks on data to go deeper
All levels always visible	Shows one level at a time
Good for letting users pick what to see	Good for guided exploration top-to-bottom
No navigation concept	Has a direction: go deeper or go back up
Simple to build	Requires hierarchy or actions to build

Module 9 — Quick Reference Cheat Sheet

Action Filters — All 3 Types

Action Type	How to Create	Run On	Best Use
Filter Action	Dashboard → Actions → Add Action → FilterSet Source Sheet, Target Sheet, Run On: Select	Click (Select)	Master-detail: click region → see its orders
Highlight Action	Dashboard → Actions → Add Action → HighlightSet Source, Target, Run On: Hover	Hover	Cross-highlight: hover over item → matching marks glow
URL Action	Dashboard → Actions → Add Action → Go to URLSet URL with <FieldName> placeholders	Menu (right-click)	Open external link related to clicked data

LOD Expressions — Formula Reference

FIXED: { FIXED [Customer] : SUM([Sales]) }
→ Total Sales per Customer, locked to Customer level
→ Ignores dimension filters on the view (use Context Filter to fix this)

INCLUDE: { INCLUDE [Order_ID] : SUM([Sales]) }
→ Calculate per Order_ID, then aggregate up to view level
→ Use AVG({INCLUDE...}) to get average order value per customer

EXCLUDE: { EXCLUDE [City] : AVG([Sales]) }
→ Remove City from the calculation – give Region-level average
→ Even if City is on the Rows shelf, it is excluded from this calc

Syntax: { LOD_TYPE [dimension(s)] : aggregation([measure]) }
Always use curly braces { } – this tells Tableau it is an LOD expression

Drill Down Methods — Comparison

Method 1: Hierarchy (+ / – Buttons)	Method 2: Action-Based Navigation
Build hierarchy in Data panel	Build separate sheets per level

Method 1: Hierarchy (+ / – Buttons)	Method 2: Action-Based Navigation
Drag parent field to chart	Add Filter Actions between sheets
+ button appears automatically	Use Show/Hide to control visibility
User clicks + to expand, – to collapse	User clicks bar to drill, button to go back
Best for: simple, quick drill down	Best for: clean professional dashboards
Complexity: Low	Complexity: Medium-High

All 3 Topics — Quick Summary

Feature	What It Does	Key Steps	Where to Find It
Filter Action	Chart A click filters Chart B	Dashboard → Actions → Add FilterSet Source and Target sheets	Dashboard menu → Actions
Highlight Action	Hover on Chart A glows Chart B	Dashboard → Actions → Add HighlightSet Run On: Hover	Dashboard menu → Actions
FIXED LOD	Calculate totals at a fixed level	Create Calc Field: { FIXED [dim] : AGG([measure]) }	Analysis → Create Calculated Field
INCLUDE LOD	Go deeper than the view	Create Calc Field: { INCLUDE [dim] : AGG([measure]) }	Analysis → Create Calculated Field
EXCLUDE LOD	Go higher than the view	Create Calc Field: { EXCLUDE [dim] : AGG([measure]) }	Analysis → Create Calculated Field
Hierarchy Drill	+ / – buttons on chart	Data panel: drag child onto parent → create hierarchy	Data panel → drag and drop fields
Action Drill Down	Click to navigate levels	Build sheets per level + Filter Actions + Show/Hide	Dashboard → Actions + Layout

End of Module 9 — Action Filters, LOD Expressions & Drill Down Reports

Practice: Build the master-detail dashboard first. Then try FIXED LOD. Finally build the hierarchy drill down.

Next → Module 10: Dashboards in Tableau — Layout, Design, Actions & Publishing